BONAP Data Aggregation Summary

C:\Users\buckw\Downloads\output_2
Generated 2026-10-24 23:49

At a glance

30,978	331	3,625	3,031	5,125	6
species / taxa	families	genera	counties scored	all-zero (16.5%)	failure-log lines

1. Data integrity

Check	Result	Status
Matrix and full file row counts match	30,978 vs 30,978	PASS
Both files list the same species in the same order	yes	PASS
Duplicate species rows	0	PASS
County columns in matrix	3,031	INFO
All-zero species (no county in grid)	5,125 (16.5%)	INFO
Species/genus/family-level scrape errors	0	PASS

Blank attribute values

Column	Blank rows
Major Plant Group	3
Habit	25
Duration	19
Nativity	1
Flower Color	9,597
Flowering Season	12,740
Habitat	15,359
Global Rank	9,220
Rarity & Endemism	4,460
Special Locations	6,623

2. Species with no county records (all-zero)

An all-zero row means no county in your coordinate grid was colored on that species' map. That is either a species that lives only outside your grid (for example Hawaii, Puerto Rico or non-US range) or a species BONAP has no map for. It is not evidence of absence.

By Nativity

Nativity	Zero rate	Zero species	Total
Continental > Native	17.7%	4,180	23,614
Continental > Exotic	17.7%	871	4,913
Continental > Native > State/Province-level > Adventive (a native plant found outside of its natural range)	2.3%	53	2,341
Continental > ?Native	64.3%	9	14
Continental > ?Exotic	38.1%	8	21
Continental > Native > Exotic	3.8%	2	53
Continental > Exotic > Hybrid Parents > One parent is native, one parent is exotic	100.0%	1	1
Continental > ?Native > State/Province-level > Adventive (a native plant found outside of its natural range)	0.0%	0	7
Continental > Exotic > State/Province-level > Adventive (in at least one state/province)	0.0%	0	1
Continental > Exotic > State/Province-level > Adventive (a native plant found outside of its natural range)	0.0%	0	4

By major plant group

Major plant group	Zero rate	Zero species	Total
Flowering plant > Dicot flowering plant	15.0%	3,610	24,058
Flowering plant > Monocot flowering plant	17.0%	931	5,480
Non-flowering plant > Pteridophyte	46.3%	559	1,207
Non-flowering plant > Gymnosperm	10.4%	24	230

Top families and genera by zero count

Family	Zero species	Total species	Zero rate
ASTERACEAE	370	3,933	9.4%
POACEAE	295	1,945	15.2%
FABACEAE	283	2,536	11.2%
CYPERACEAE	246	1,238	19.9%
CAMPANULACEAE	191	347	55.0%
RUBIACEAE	169	424	39.9%
ORCHIDACEAE	134	414	32.4%
LAMIACEAE	118	742	15.9%
GESNERIACEAE	118	118	100.0%
MYRTACEAE	117	163	71.8%

Genus	Zero species
Carex	106
Cyrtandra	104
Cyanea	96
Asplenium	57
Salix	52
Melicope	49
Euphorbia	49
Dubautia	49
Peperomia	45
Psychotria	42

Zero streaks in file order

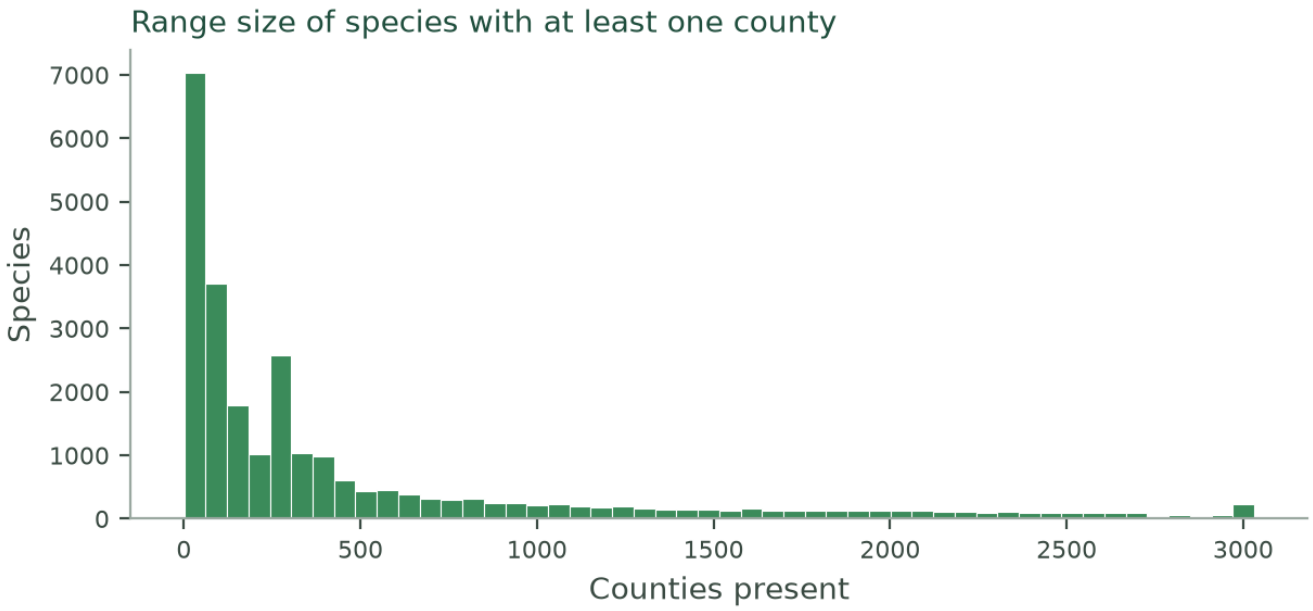
Longest run of consecutive all-zero rows: 153. Clustered zeros usually mean one region-specific genus; zeros scattered across unrelated families would point to a timing problem.

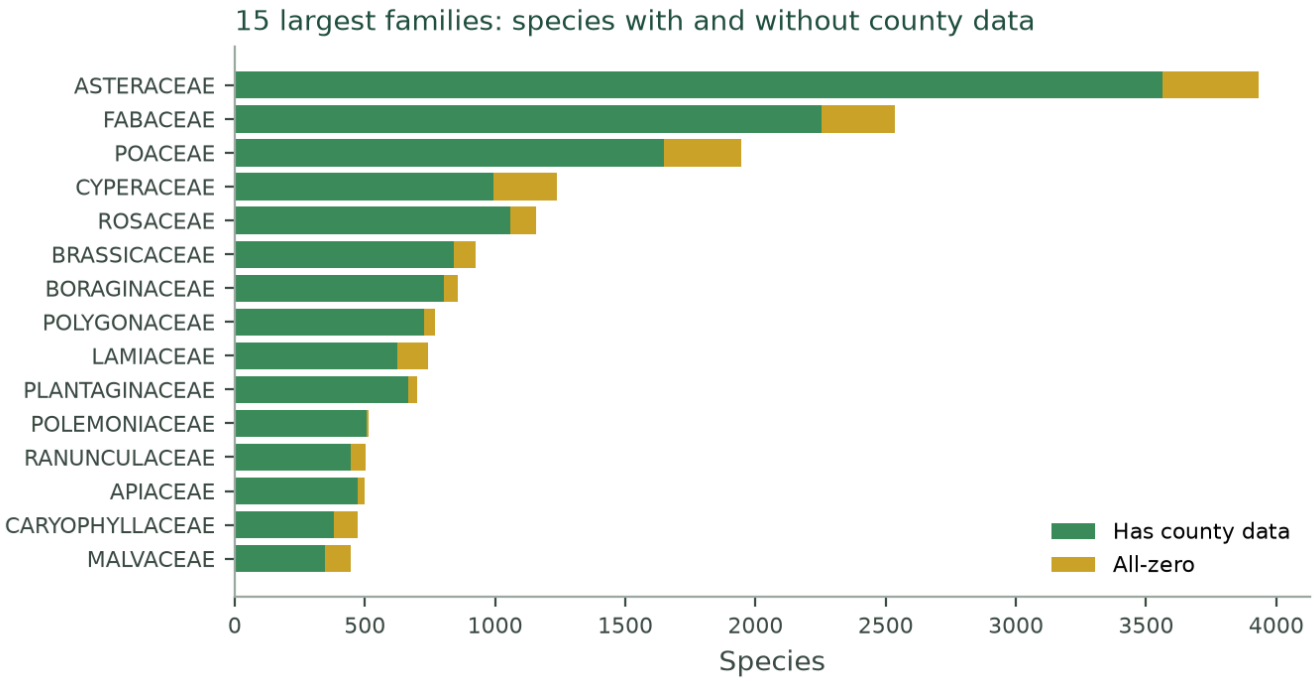
Family	Genus	Species in run
CAMPANULACEAE	Cyanea	96
CAMPANULACEAE	Clermontia	40
CAMPANULACEAE	Delissea	17

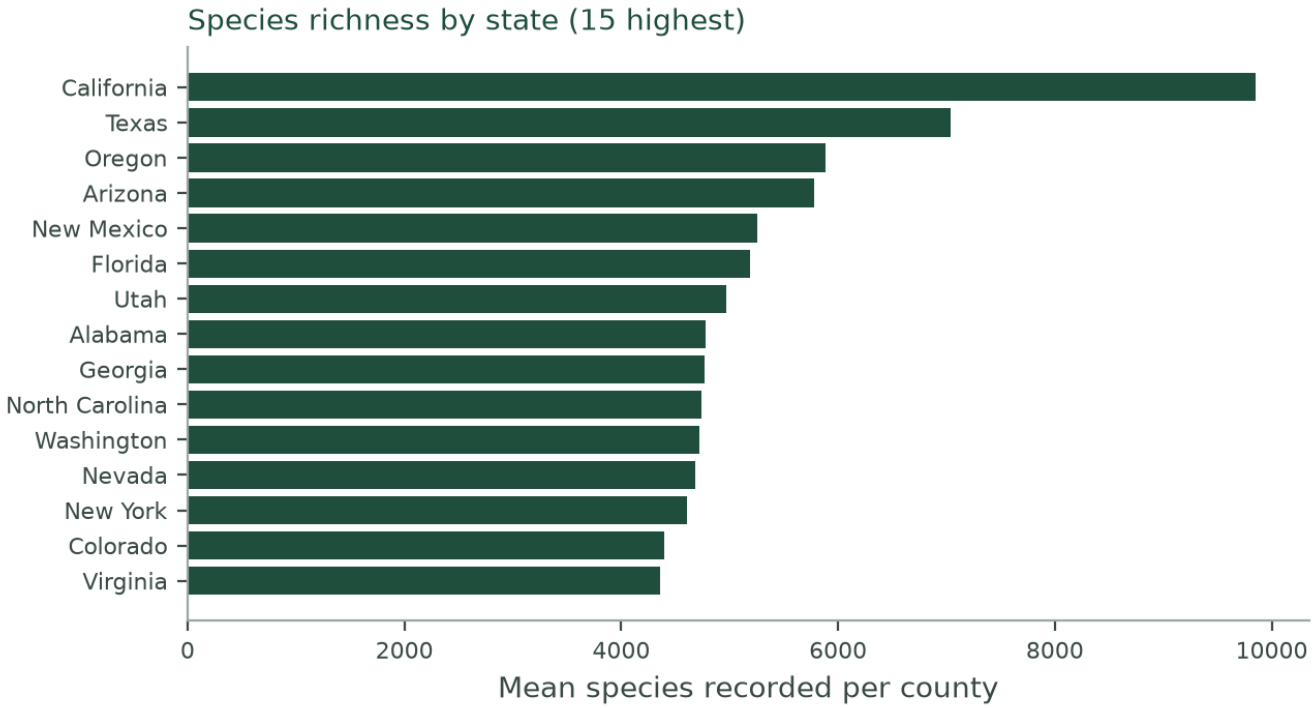
15 random zero species to spot-check on bonap.net

Family	Genus	Species	Nativity
MYRTACEAE	Leptospermum	*Leptospermum petersonii*	Continental > Exotic
CARYOPHYLLACEAE	Arenaria	*Arenaria pseudofrigida*	Continental > Native
ARECACEAE	Pritchardia	*Pritchardia hillebrandii*	Continental > Native
ORCHIDACEAE	Microchilus	*Microchilus plantagineus*	Continental > Native
CAMPANULACEAE	Cyanea	*Cyanea solenocalyx*	Continental > Native
PIPERACEAE	Piper	*Piper betle*	Continental > Exotic
ARACEAE	Anthurium	*Anthurium dominicense*	Continental > Native
BORAGINACEAE	Cordia	*Cordia obliqua*	Continental > Exotic
POACEAE	Euclasta	*Euclasta condylotricha*	Continental > Native
CAMPANULACEAE	Cyanea	*Cyanea mannii*	Continental > Native
ORCHIDACEAE	Arundina	*Arundina graminifolia*	Continental > Exotic
RUTACEAE	Melicope	*Melicope molokaiensis*	Continental > Native
RUBIACEAE	Catesbaea	*Catesbaea melanocarpa*	Continental > Native
ACANTHACEAE	Dicliptera	*Dicliptera chinensis*	Continental > Exotic
CONVOLVULACEAE	Cuscuta	*Cuscuta globulosa*	Continental > Native

3. Charts







4. County coverage by state

State	Counties	State	Counties	State	Counties	State	Counties
Alabama	67	Iowa	99	Nevada	17	South Dakota	66
Arizona	15	Kansas	105	New Hampshire	10	Tennessee	95
Arkansas	75	Kentucky	119	New Jersey	20	Texas	254

State	Counties	State	Counties	State	Counties	State	Counties
California	58	Louisiana	64	New Mexico	33	Utah	29
Colorado	63	Maine	14	New York	59	Vermont	13
Connecticut	8	Maryland	23	North Carolina	99	Virginia	94
Delaware	3	Massachusetts	11	North Dakota	53	Washington	37
District of Columbia	1	Michigan	83	Ohio	87	West Virginia	55
Florida	54	Minnesota	87	Oklahoma	77	Wisconsin	72
Georgia	156	Mississippi	82	Oregon	36	Wyoming	23
Idaho	44	Missouri	114	Pennsylvania	66		
Illinois	102	Montana	56	Rhode Island	2		
Indiana	92	Nebraska	93	South Carolina	46		

Not in the coordinate file: HI, AK, PR, VI. Species found only there will appear as all-zero.

5. Failure logs

6 log file(s), 6 line(s). 0 species/genus/family-level error(s); 6 species logged as 'nomap' (no map image retrieved, recorded as all zeros).

Level	Lines
nomap	6

Level	Family	Genus	Species	Error
nomap	CUPRESSACEAE	Callitris	*Callitris endlicheri*	map not retrieved
nomap	FABACEAE	Mimosa	*Mimosa pigra*	map not retrieved
nomap	POACEAE	Poa	*Poa pachypholis*	map not retrieved
nomap	GOODENIACEAE	Scaevola	*Scaevola taccada*	map not retrieved
nomap	CONVOLVULACEAE	Cuscuta	*Cuscuta chinensis*	map not retrieved
nomap	CONVOLVULACEAE	Cuscuta	*Cuscuta chinensis var. applanata*	map not retrieved

6. Varieties and subspecies vs. parent species

6,794 infraspecific taxa have a parent species in the data. 2,682 (39.5%) have a county pattern identical to the parent (2,154 of 6,003 when all-zero rows are excluded). A high share suggests BONAP displays the parent map for those taxa.

7. Range size summary

Statistic	Value
Species with data	25,853.0
Mean counties	494.6
Std. dev.	686.4
Minimum	1.0
25th percentile	57.0
Median	203.0
75th percentile	568.0
Maximum	3,031.0

Widest-ranging species	Counties
Bidens frondosa	3,031
Typha latifolia	3,031
Plantago lanceolata	3,031
Poa pratensis ssp. pratensis	3,031
Taraxacum officinale	3,031
Rumex acetosella	3,031
Poa pratensis	3,031
Cerastium fontanum ssp. vulgare	3,031
Cerastium fontanum	3,031
Trifolium repens	3,031

Most species-rich counties	Species
Amador, California	9,947
Marin, California	9,946
Lake, California	9,946
Kings, California	9,946
Imperial, California	9,946
Butte, California	9,946
El Dorado, California	9,946
Glenn, California	9,946
San Francisco, California	9,946
Santa Cruz, California	9,946

Counties with zero species recorded: 0.